

Three Macro Models for Fluctuations and Monetary Non-neutrality

A compact, exam-style summary of: *Real Business Cycle*, *Money-in-the-Utility*, and *Menu Cost*.

Real Business Cycle (RBC) Model

Goal. Explain business-cycle fluctuations using *real* shocks (technology), with competitive markets and intertemporal household optimization. Technology shocks move output and labor through optimal responses.

Core ingredients (discrete time).

- Production:

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}, \quad 0 < \alpha < 1.$$

- Technology shock (stochastic productivity):

$$\ln A_t = \rho_A \ln A_{t-1} + \varepsilon_t.$$

- Household optimality conditions summarized by:

$$\begin{aligned} \text{Intratemporal labor supply: } \frac{b}{1-L_t} &= \frac{W_t}{C_t}, \\ \text{Euler equation: } \frac{1}{C_t} &= e^{-\rho} E_t \left[\frac{1}{C_{t+1}} (1 - \delta + r_{t+1}) \right]. \end{aligned}$$

What to remember.

- *Driver of fluctuations:* ε_t (technology shock) shifts marginal products, hence r_t, W_t , and optimal (C_t, L_t) adjustments.
- *Solution approach:* typically requires linearization (log-linear system) and then simulation/IRFs.
- *Limitation emphasized in the notes:* omits monetary shocks, so it struggles to match evidence of monetary effects.

Money-in-the-Utility (MIU) Model with Fixed Prices

Goal. Show how monetary shocks can have *real* effects when there is *nominal rigidity* (prices fixed). Money enters utility through real balances M_t/P_t .

Core ingredients.

- Household utility includes real money balances (MIU) and labor disutility; households choose (C_t, M_t, L_t) subject to a wealth evolution constraint.
- Euler equation implies an “IS” relation (written in real terms). Under equilibrium $Y_t = C_t$, this yields a New Keynesian IS-type equation relating Y_t and the real rate.
- Money demand plus *fixed prices* $P_t = \bar{P}$ gives an “LM” relation:

$$\log Y_t = \log Y_{t+1} - \frac{1}{\theta} \log[(1 + r_t)\beta], \quad \frac{M_t}{\bar{P}} = Y_t^{\theta/\nu} \left(\frac{1}{r_t} + 1 \right)^{1/\nu}.$$

Mechanism (fixed prices).

- Increasing money supply shifts LM and lowers r_t , raising Y_t in the IS–LM diagram; hence money has real effects *given nominal rigidity*.
- Key caveat in the notes: with fully competitive goods and labor markets, fixed prices alone cannot explain why firms supply the extra output (requires some imperfection).

Menu Cost Model (Imperfect Competition + Price Adjustment Cost)

Goal. Microfound nominal rigidity: prices are sticky because changing them has a fixed cost (menu cost). With imperfect competition, sticky prices make monetary shocks affect real output.

Static baseline (continuum of monopolistic firms).

- Differentiated goods and CES aggregation imply demand:

$$Y(i) = \left(\frac{P(i)}{P} \right)^{-\eta} Y, \quad \eta > 1.$$

- Monopoly pricing yields a constant markup over marginal cost:

$$\frac{P(i)}{P} = \frac{\eta}{\eta - 1} \frac{W}{P}.$$

- Money enters via the quantity equation:

$$MV = PY.$$

Monetary shock with menu costs.

- After a money increase $M \rightarrow M'$, each firm compares the benefit of repricing (higher profit at the new optimal price) to the fixed menu cost Z .
- If the menu cost is large enough (in the notes: $Z > \pi_A - \pi_F$), firms do not adjust prices; the aggregate price level stays fixed, so the quantity equation implies output rises proportionally with money:

$$Y : \frac{MV}{P} \rightarrow \frac{M'V}{P}.$$

One-Page Comparison Table

Feature	RBC Model	MIU Model (Fixed Prices)	Menu Cost Model
Main shock / driver	Technology shock ε_t in A_t	Money supply shift M_t with $P_t = \bar{P}$	Money supply shift $M \rightarrow M'$ plus menu cost Z
Nominal rigidity	Absent (prices flexible)	Imposed exogenously (prices fixed)	Endogenized via menu cost Z
Key equilibrium objects	Euler + labor supply + tech process	IS + LM under $P_t = \bar{P}$	Markup pricing + quantity equation + (no) repricing condition
Why money affects output?	It does not (by construction; “real” model)	With fixed prices, LM shifts change r and Y	Sticky prices from Z keep P fixed so $MV = PY$ forces Y up
Key limitation (as stated)	Omits monetary shocks	Needs market imperfection to justify supply response	Static baseline; relies on imperfect competition and adjustment costs